

Exp No: 16

SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

OBJECTIVES

- To configure the UART serial communication in the AT89C51 microcontroller.
- To enable and handle serial interrupts for keyboard input.
- To display “Hello World” continuously and display an interrupt message when a key is pressed.

PROGRAM

```
#include <reg51.h>

#include <stdio.h>

void serial_ISR(void) interrupt 4
{
    char ch;

    if (RI)
    {
        RI = 0; // Clear receive interrupt flag
        ch = SBUF; // Read received character
        printf("\n>>> INTERRUPT RECEIVED: ");
        printf("%c\n", ch);
    }
}

void delay(void)
{
    unsigned int i;

    for(i = 0; i < 50000; i++); // Delay for visibility
}

void main(void)
{

```

```
SCON = 0x50; // UART mode 1, 8-bit, REN enabled
```

```
TMOD = 0x20; // Timer1 mode 2 (auto-reload)
```

```
TH1 = 0xFD; // Baud rate 9600
```

```
TR1 = 1; // Start Timer1
```

```
IE = 0x90; // EA = 1, ES = 1 (enable serial interrupt)
```

```
TI = 1; // Enable printf
```

```
while(1)
```

```
{
```

```
printf("Hello World\n");
```

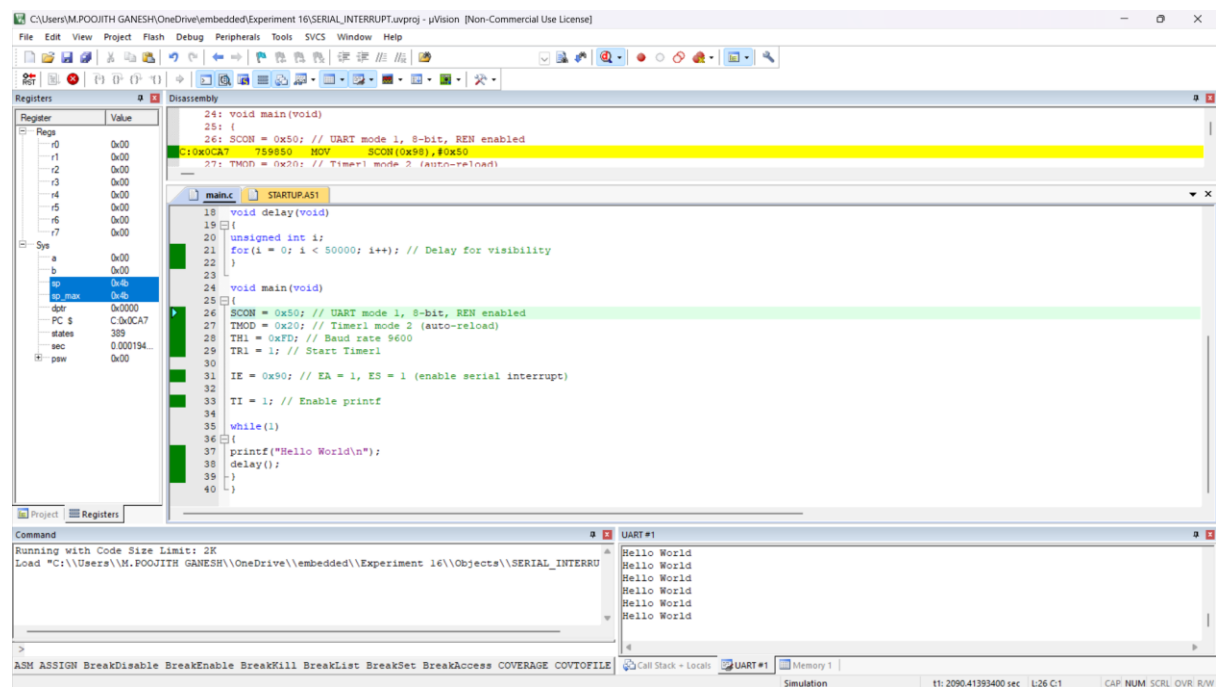
```
delay();
```

```
}
```

```
}
```

OUTPUT

a) Displaying a Message:



b) During interruption:

The screenshot shows an IDE with the following components:

- Registers:** A table showing the state of registers. The PC register is highlighted with the value 0x0CA7.
- Disassembly:** A window showing the assembly code for the program. The main function is highlighted, showing the initialization of the UART and the delay loop.
- main.c:** A window showing the C source code. The main function is highlighted, showing the initialization of the UART and the delay loop.
- Command:** A window showing the output of the program. The output is "Hello World" followed by "Interrupt received: a" and "Hello World" repeated three times.
- UART #1:** A window showing the UART data. The output is "Hello World" repeated three times.

```
24: void main(void)
25: {
26:     SCON = 0x50; // UART mode 1, 8-bit, REN enabled
27:     TMOD = 0x20; // Timer1 mode 2 (auto-reload)
28:     TH1 = 0xFD; // Baud rate 9600
29:     TR1 = 1; // Start Timer1
30:
31:     IE = 0x90; // EA = 1, ES = 1 (enable serial interrupt)
32:     TI = 1; // Enable printf
33:
34:     while(1)
35:     {
36:         printf("Hello World\n");
37:         delay();
38:     }
39: }
40: }
```

Running with Code Size Limit: 2K
Load "C:\Users\H.M. POOJITH GANESH\OneDrive\embedded\Experiment 16\Objects\SERIAL_INTERRUPT"

UART #1

Hello World

>>> INTERRUPT RECEIVED: a

Hello World

Hello World

Hello World

Simulation

11: 2548.39350850 sec L26 C1 CAP. NUM SCRL OVR. R/W